KRISHNA PRANEET GUDIPATY

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RESEARCH INTERESTS

Software Engineering, Distributed Systems, Model Serving and Placement, Cloud and Edge resource management, Failure resilience and recovery, Foundation Models and multimodal AI

EDUCATION

University of Massachusetts Amherst (UMass Amherst) **Amherst, MA**
Ph.D., Computer Science, (GPA: 4.0/4.0), Advisor: Prashant Shenoy *Jul 2024 – May 2028 (expected)*
M.S., Computer Science, (GPA: 3.97/4.0), Advisor: Stefan Krastanov *Sep 2022 – May 2024*

- **Relevant Coursework**: Advanced Natural Language Processing, Machine Learning, Secure Distributed Systems, Advanced Algorithms, Systems for Machine Learning, Distributed & Operating Systems, Reinforcement Learning, Software Engineering

Indian Institute of Technology Madras (IITM) **Chennai, India**
B.Tech., Metallurgical & Materials Engg., (GPA: 8.75/10), Advisor: Satyesh Kumar Yadav *Aug 2016 – Dec 2020*

- **Relevant Coursework**: Cloud Computing, C Programming, Probability & Statistics, Differential Equations, Linear Algebra, Computational Materials

EXPERIENCE

University of Massachusetts Amherst, Research Assistant *Jul 2024 – Present*
Resilient Foundation Model (FM) serving using structural model stitching (ongoing)

- Developing FLOPs-effective redundancy techniques for failure-resilient Foundation Model serving at the edge, achieving graceful degradation with **5%** accuracy-loss with only **4%** memory overheads
- Proposed a system utilizing an online primal-dual pricing and placement algorithm to balance rapid failover with resilience to co-located and dependent failures

Foundation Models (FMs) for the Edge: Opportunities and Trade-offs, paper

- Evaluated post-training techniques with LoRA and linear probing to reduce adaptation and quantization costs across vision and time-series FMs for edge and resource-constrained IoT deployments
- Characterized downstream task adaptations, showcasing cases where fine-tuning data and training FLOPs can be reduced by **90%** and **95%** respectively, compared to traditional supervised models

Hydra: Multi-Level Ensemble Learning for Failure-Resilient Edge Inference, paper

- Designed a distributed hierarchical ensemble model over gRPC to provide multi-fault tolerance and reduce resource costs by **40%**, and evaluated the approach across CNNs, RNNs, Vision Transformers, and LLMs
- Improved failure-mode latency by **25%** compared to baselines while retaining **95%** of the original model accuracy

Practical Considerations for Failure Resilient ML Systems at the Edge, paper

- Conducted a systems study of fault-tolerance overheads in current designs across enterprise and edge GPUs
- Proposed directions for improving failure diagnostics, localization, and multi-agent consensus controllers

Center for Quantum Networks (CQN), Software Engineer Intern *Jan 2024 – May 2024*
Distributed Quantum Network Simulator

- Designed a REST API server using Oxygen.jl to enable programmatic access to a quantum network simulator
- Integrated HTTP endpoints for manipulating qubit registers and gates across nodes in the network using QuantumSavory.jl, enabling reproducible research in quantum coding theory

Quantum Information Lab, Research Assistant *Jun 2023 – May 2024*
Efficient implementation of LDPC decoders for error correction, code

- Developed LDPCDecoders.jl, an open source Julia library for quantum error correction, **500+** JuliaHub downloads
- Reduced decoding latency by **30%** over baselines (python-qldpc) and released the first iterative decoder
- Integrated benchmarking API endpoints into a quantum simulator in under **10** lines of code for reproducibility

Deskera, Software Engineer *Dec 2020 – Jul 2022*

Distributed microservices and event-driven data pipelines

- Implemented Java microservices automating e-commerce workflows, cutting manual operational overhead by **40%**
- Built event-driven pipelines with Kafka and webhooks, syncing data across POS systems and web stores
- Designed React UI pages for configuration management, KPI tracking, and invoicing in Deskera's ERP software

Indian Institute of Technology Madras, Undergraduate Researcher *Aug 2019 – May 2020*

Machine Learning for accelerating high-performance and parallel AIMD simulations

- Experimented with non-linear ML models over atomic positions to accelerate Platinum molecular dynamics
- Achieved **1–5%** MAE with Gaussian Kernel Ridge Regression on unseen data from different structural environments, assessing the generalizability of the ML force fields
- Implemented *numerical fingerprinting* from theory, devising an algorithm using rotation and kernel matrices on spherical atomic coordinates to prepare input data generated with VASP on an IBM supercluster

Publicis Sapient, Software Engineer Intern *Jun 2019 – Jul 2019*

Latency-sensitive and strictly consistent distributed trading platform

- Optimized the Java trading backend, improving server utilization and cutting order book latency by **25%**
- Implemented cross-domain login using OAuth 2.0 and revamped the frontend for the portfolio service using React

Boeing, Software Research Intern *May 2018 – Jul 2018*

Real-time defect recognition algorithms for Non-Destructive Evaluation

- Developed image processing pipelines for defect detection and to quantify shrinkage in aircraft components, achieving a segmentation accuracy of **95%**
- Analyzed industrial DICOM images using frequency-domain filtering (Gaussian, Butterworth high-pass) and morphological operations (dilation, erosion)

PUBLICATIONS

- **Short Paper: Hydra: Multi-Level Ensemble Learning for Failure-Resilient Edge Inference**
  Gudipaty, K. P., Hanafy, W. A., Ng, N., Ozkara, K., Liang, Q., Milzman, J., Irwin, D., Diggavi, S., & Shenoy, P.
  In Proceedings of the Eleventh ACM/IEEE Symposium on Edge Computing (SEC), 2026.
- **Foundation Models for the Edge: Opportunities and Trade-offs**
  Gudipaty, K. P., Hanafy, W. A., Abdelzaher, T., Diggavi, S., Irwin, D., Srivastava, M., & Shenoy, P.
  In Proceedings of IEEE Military Communications Conference (MILCOM), 2026.
- **Practical Considerations for Failure Resilient ML Systems at the Edge**
  Gudipaty, K. P., Hanafy, W. A., Wu, L., Twigg, J., Marlin, B. M., Milzman, J., Diggavi, S., Abdelzaher, T., & Shenoy, P.
  In Proceedings of IEEE Military Communications Conference (MILCOM), 2025.
- **Quantum LDPC Error Correcting Codes for Use on 1D Quantum Dot Arrays**
  Micciche, A., **Gudipaty, K. P.**, & Krastanov, S.
  Abstract and talk, *APS March Meeting*, 2024.

SKILLS

Languages & Frameworks: C/C++, Java, Python, MATLAB, Go, Julia, HTML, CSS, React, TypeScript, SQL
Machine Learning: PyTorch, TensorFlow, Pandas, NumPy, scikit-learn, Spark, Matplotlib
Infrastructure & Tooling: GCP, AWS, Linux, Docker, Kubernetes, Kafka, Grafana, Jenkins, Postgres, MongoDB

ACTIVITIES AND SERVICES

Awards: ACM/IEEE SEC 2026 Student Travel Grant
Teaching Assistant: CS365 Digital Forensics, CS677 Distributed and Operating Systems, CS230 Computer Systems,
Mentoring: UMass Turing Program 2026 (Guest Lecturer), URV Program 2026, LASS Lab Research 2025
Reviewing: *NeurIPS* 2026, *AISTATS* 2026, *OSDI* 2026 AE Committee, *MLSys* 2026 AE Committee
Student Volunteering: Student Systems Lunch Seminar, UMass Amherst (2025)